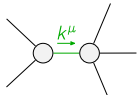


$$\begin{array}{c}
\begin{array}{cc}
\mathcal{K}_{0^+}^{\#1} & \mathcal{K}_{0^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{K}_{0^+}^{\#1}{}^\dagger & \begin{array}{|c|c|} \hline -\frac{3k^2{}^{(4)}\kappa_1}{2} & 0 \\ \hline \end{array} \\
\mathcal{K}_{0^+}^{\#1}{}^{\alpha\beta\chi} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array}
\end{array}
\begin{array}{cccc}
\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta} & \mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta} & \mathcal{K}_{1^+}^{\#1}{}_{\alpha} & \mathcal{K}_{1^+}^{\#2}{}_{\alpha} \\
\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta} & \begin{array}{|c|c|} \hline -k^2{}^{(4)}\kappa_3 & -\frac{k^2{}^{(4)}\kappa_3}{\sqrt{2}} \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta} & \begin{array}{|c|c|} \hline -\frac{k^2{}^{(4)}\kappa_3}{\sqrt{2}} & -\frac{k^2{}^{(4)}\kappa_3}{2} \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array}
\end{array}
\begin{array}{cc}
\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta} & \mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{K}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{2^+}^{\#1}{}^{\alpha\beta\chi} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array}
\end{array}
\end{array}$$

$$\begin{array}{c}
\begin{array}{cc}
\mathcal{J}_{0^+}^{\#1} & \mathcal{J}_{0^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{J}_{0^+}^{\#1}{}^\dagger & \begin{array}{|c|c|} \hline -\frac{2}{3k^2{}^{(4)}\kappa_1} & 0 \\ \hline \end{array} \\
\mathcal{J}_{0^+}^{\#1}{}^{\alpha\beta\chi} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array}
\end{array}
\begin{array}{cccc}
\mathcal{J}_{1^+}^{\#1}{}_{\alpha\beta} & \mathcal{J}_{1^+}^{\#2}{}_{\alpha\beta} & \mathcal{J}_{1^+}^{\#1}{}_{\alpha} & \mathcal{J}_{1^+}^{\#2}{}_{\alpha} \\
\mathcal{J}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta} & \begin{array}{|c|c|} \hline -\frac{4}{9k^2{}^{(4)}\kappa_3} & -\frac{2\sqrt{2}}{9k^2{}^{(4)}\kappa_3} \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta} & \begin{array}{|c|c|} \hline -\frac{2\sqrt{2}}{9k^2{}^{(4)}\kappa_3} & -\frac{2}{9k^2{}^{(4)}\kappa_3} \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#1}{}^\dagger{}^{\alpha} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#2}{}^\dagger{}^{\alpha} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array}
\end{array}
\begin{array}{cc}
\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta} & \mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{J}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{2^+}^{\#1}{}^{\alpha\beta\chi} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array}
\end{array}
\end{array}$$

Lagrangian

$$\kappa_1^{(4)}\partial_\beta\mathcal{K}^\delta{}_{\chi\delta}\partial^\chi\mathcal{K}^\alpha{}_\alpha{}^\beta + \kappa_3^{(4)}\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^\delta{}_{\alpha\chi} - 2\kappa_3^{(4)}\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^\delta{}_{\beta\chi} - \kappa_3^{(4)}\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^\delta{}_{\chi\alpha} - \frac{1}{2}\kappa_3^{(4)}\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^\delta{}_{\beta\chi}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{0^+}^{\#1\alpha\beta\chi} == 0$	1	$\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + \partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + \partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + \partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} ==$ $\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + \partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + \partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}$	
$\mathcal{J}_{1^+}^{\#2\alpha} == 0$	3	$\partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi} == 0$	
$\mathcal{J}_{1^+}^{\#1\alpha} == 0$	3	$\partial_\chi\partial^\alpha\mathcal{J}^\beta{}_\beta{}^\chi + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_\beta == \partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi}$	
$\mathcal{J}_{1^+}^{\#1\alpha\beta} == \mathcal{J}_{1^+}^{\#2\alpha\beta}$	3	$3\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi} + 2\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\chi\alpha\beta} == 3\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi}$	
$\mathcal{J}_{2^+}^{\#1\alpha\beta\chi} == 0$	5	$3\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon} + 3\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_\delta + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + 4\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} +$ $4\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + 3\eta^{\beta\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^\delta{}_\delta{}^\epsilon + 3\eta^{\alpha\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon} + 3\eta^{\beta\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha}{}_\delta ==$ $3\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon} + 3\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_\delta + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + 4\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} +$ $2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} + 4\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta} + 3\eta^{\alpha\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^\delta{}_\delta{}^\epsilon + 3\eta^{\beta\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon} + 3\eta^{\alpha\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta}{}_\delta$	
$\mathcal{J}_{2^+}^{\#1\alpha\beta} == 0$	5	$3\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 3\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\eta^{\alpha\beta}\partial_\epsilon\partial^\epsilon\partial_\delta\mathcal{J}^\chi{}_\chi{}^\delta == 2\partial_\delta\partial^\beta\partial^\alpha\mathcal{J}^\chi{}_\chi{}^\delta + 3(\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi})$	
Total # constraint(s):	20		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$-\frac{1}{\kappa_3^{(4)}}$

Resolved unitarity condition(s):

$$\kappa_3^{(4)} < 0$$